SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1515

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct

I ***KESHAVARAM L/192511253***(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

L. Kesh

Signature of the student

Assessment Tasks

Application Based Questions

Q1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Answer:

Auto-scaling automatically adds or removes cloud instances according to demand, while load balancing distributes incoming requests across healthy instances. During a seasonal sale, extra application servers can be launched as traffic increases and removed when traffic falls.

This prevents overloaded servers, reduces response time, and improves availability. Monitoring tools track CPU, memory, request rate, latency, error rate, and server health; their metrics trigger scaling rules and alerts.

For example, an e-commerce website during a festival sale can add servers when thousands of customers browse and purchase products. The load balancer spreads requests across them, helping users search, add items to carts, and complete payments with fewer delays.

Q2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Answer:

Cloud object storage can replicate data across multiple devices or availability zones to improve durability and redundancy. Versioning, lifecycle policies, and geographically separate copies provide additional recovery options.

Encryption protects data at rest and in transit. Identity and access management, least-privilege permissions, multi-factor authentication, and immutable backup policies reduce unauthorized access or deletion.

Cloud storage can reduce upfront hardware purchases, maintenance, power, and physical-space costs. Organizations can pay for the capacity they use and move older backups to lower-cost archival tiers.

For example, if a company's local server room is damaged, it can restore databases and files from encrypted cloud backups in another region and resume operations.

Q3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Answer:

A CDN caches static content such as images, scripts, and video segments at edge servers close to users. Requests are served from a nearby edge location instead of travelling to the origin server, reducing latency and origin load.

Cloud providers offer global regions, availability zones, load balancing, and health checks to support resilient deployments. Dynamic application services can be deployed in multiple regions where required.

For example, a global video-streaming app can deliver popular video segments from local edge caches, reducing buffering. A game can use regional servers to reduce round-trip time, improving responsiveness for players in different countries.

Q4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Answer:

Cloud providers offer compliance certifications, audit reports, policy tools, and services that help customers meet applicable requirements. The institution remains responsible for configuring services correctly and meeting its own legal obligations.

Logs record access, administrative actions, and transactions. Monitoring detects unusual activity and alerts security teams. Encryption protects data in transit and at rest; key management limits who can decrypt it.

To meet data sovereignty requirements, select approved regions, restrict replication and backups to permitted locations, document data flows, and review provider terms and applicable laws.

For example, a bank can process payments in an approved region, encrypt transaction records, log administrator access, and alert on unusual access or transaction patterns.

Q5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Answer:

Containers package an application and its dependencies while sharing the host operating-system kernel. They usually start quickly and use fewer resources than VMs, which each run a guest operating system on virtualized hardware. VMs provide a stronger separate OS boundary, while containers still require careful security configuration.

Kubernetes automates container deployment, service discovery, health checks, recovery, rolling updates, and scaling based on demand.

Containers built from standard images can run consistently on laptops, on-premises servers, and different cloud providers, reducing environment differences. For example, an online shopping application can run separate catalog, cart, payment, and notification microservices in containers, each scaled independently.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity